

Licredity Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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Assisting Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Licredity on Licredity. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated September 2, 2025, was delivered to Licredity, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 13 days, the Cyfrin team conducted an audit on the Licredity code provided by Licredity. The findings consist of 2 Critical, 2 High, 1 Medium & 4 Low severity issues with the remainder being informational.

Summary

Project Name	Licredity
Repository	licredity-v1-core
Commit	e8ae10a7d9f2...
Fix Commit	aff3771e437b...
Repository 2	licredity-v1-oracle
Commit	1ec4b09826b4...
Fix Commit	c387983ee0a3...
Audit Timeline	Aug 11th - Aug 27th, 2025
Methods	Manual Review, Formal Verification

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	2	2	0
High Risk	2	2	0
Medium Risk	1	1	0
Low Risk	4	3	1
Informational	13	10	3
Gas Optimizations	0	0	0
Total	22	18	4

6 Audit Scope

core:

```
src/types/Position.sol
src/types/InterestRate.sol
src/types/NonFungible.sol
src/types/Fungible.sol
src/types/FungibleState.sol
src/RiskConfigs.sol
src/libraries/PipsMath.sol
src/libraries/Locker.sol
src/libraries/ChainInfo.sol
src/libraries/FullMath.sol
src/Licredity.sol
src/BaseERC20.sol
src/Extsload.sol
src/BaseHooks.sol
```

oracle:

```
src/ChainlinkOracleConfigs.sol
src/libraries/ChainlinkFeedLibrary.sol
src/libraries/FixedPointMath.sol
src/ChainlinkOracle.sol
src/modules/uniswap/v4/types
src/modules/uniswap/v4/types/PositionInfo.sol
src/modules/uniswap/v4/UniswapV4Module.sol
src/modules/uniswap/v4/interfaces
src/modules/uniswap/v4/interfaces/IPositionManager.sol
src/modules/uniswap/v3/UniswapV3Module.sol
src/modules/uniswap/v3/libraries
src/modules/uniswap/v3/libraries/PoolAddress.sol
src/modules/uniswap/v3/interfaces
src/modules/uniswap/v3/interfaces/IUniswapV3Pool.sol
src/modules/uniswap/v3/interfaces/INonfungiblePositionManager.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Licredity is a lending protocol that utilizes a Uniswap V4 hook to enable trading on the interest rate. Each Licredity contract is a hook attached to a Uniswap pool consisting of a base token and a debt (lending) token. The debt token is backed by collateral held in the Licredity (hook) contract. Interest accrued on loans is donated to liquidity providers that are in range of the spot price at the moment of accrual. Collateral can be ERC-20 tokens and/or ERC-721 Uniswap V3 or V4 LP positions.

The protocol also uses an oracle to value collateral. This oracle combines the pool's debt/base price (smoothed via EMA) with Chainlink price feeds for the collateral assets.

Liquidity providers provide liquidity to the pool and earn donated interest in addition to trading fees, borrowers borrow the debt token against posted collateral, and traders trade on the interest price in the pool.

8.1 Key Components

- **Licredity:** Main hook and lending contract. Manages per-position collateral (ERC-20 and Uniswap v3/v4 LP NFTs), mints/burns the debt token, and accrues interest that is periodically donated to in-range LPs. The `exchange` function provides a bounded 1:1 exchange between the base and debt tokens, and liquidations (`seize`) transfer position ownership.
- **Oracle:** Returns a position's value denominated in the debt token. It blends the pool's debt/base spot price with an EMA of the previous update and values each collateral asset using Chainlink feeds.

9 Attestation

Cyfrin attests that it performed a security review of the Licredity code provided by Licredity.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-09-01-cyfrin-licredity-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Licredity has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=bce15179e07bd20f884a56455d05f484b89dcba04462c62117cd05c7b0368e70
F=2025-09-01-cyfrin-licredity-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.